

# E-Business Systems and Applications

**E-business** means using **digital technology** to perform **business activities** such as buying, selling, marketing and managing business operations.

In agriculture, e-business helps farmers, buyers and other stakeholders **trade products, manage supply chains and access information online**.

## E-Business in Agriculture



## Applications of E-Business



## **Benefits of E-Business**

- Saves **time and cost**.
- Improves market access.
- Reduces delays in the supply chain.
- Provides information quickly.
- Makes transactions easier.
- Improves transparency.
- Helps farmers make better decisions.
- Connects farmers with more buyers.

# E-Commerce System and Applications

**E-commerce (Electronic Commerce)** means **buying and selling goods and services through the internet**.

In agriculture, e-commerce connects **farmers, buyers, consumers and businesses** through online platforms.

## E-Commerce in Agriculture

Traditional agriculture often has many **middlemen** between farmers and consumers. E-commerce allows farmers to sell their products **directly to buyers**, reducing the number of intermediaries.



## Applications of E-Commerce in Agriculture



## **Benefits of E-Commerce in Agriculture**

- **Wider market:** Farmers can reach more buyers.
- **Better price information:** Farmers can compare market prices.
- **Less middlemen:** Direct selling can reduce intermediaries.
- **Lower wastage:** Better supply chain management.
- **Easy transactions:** Online buying and payment are convenient.
- **More information:** Farmers get market and farming information.
- **Better customer access:** Consumers can get fresh agricultural products.

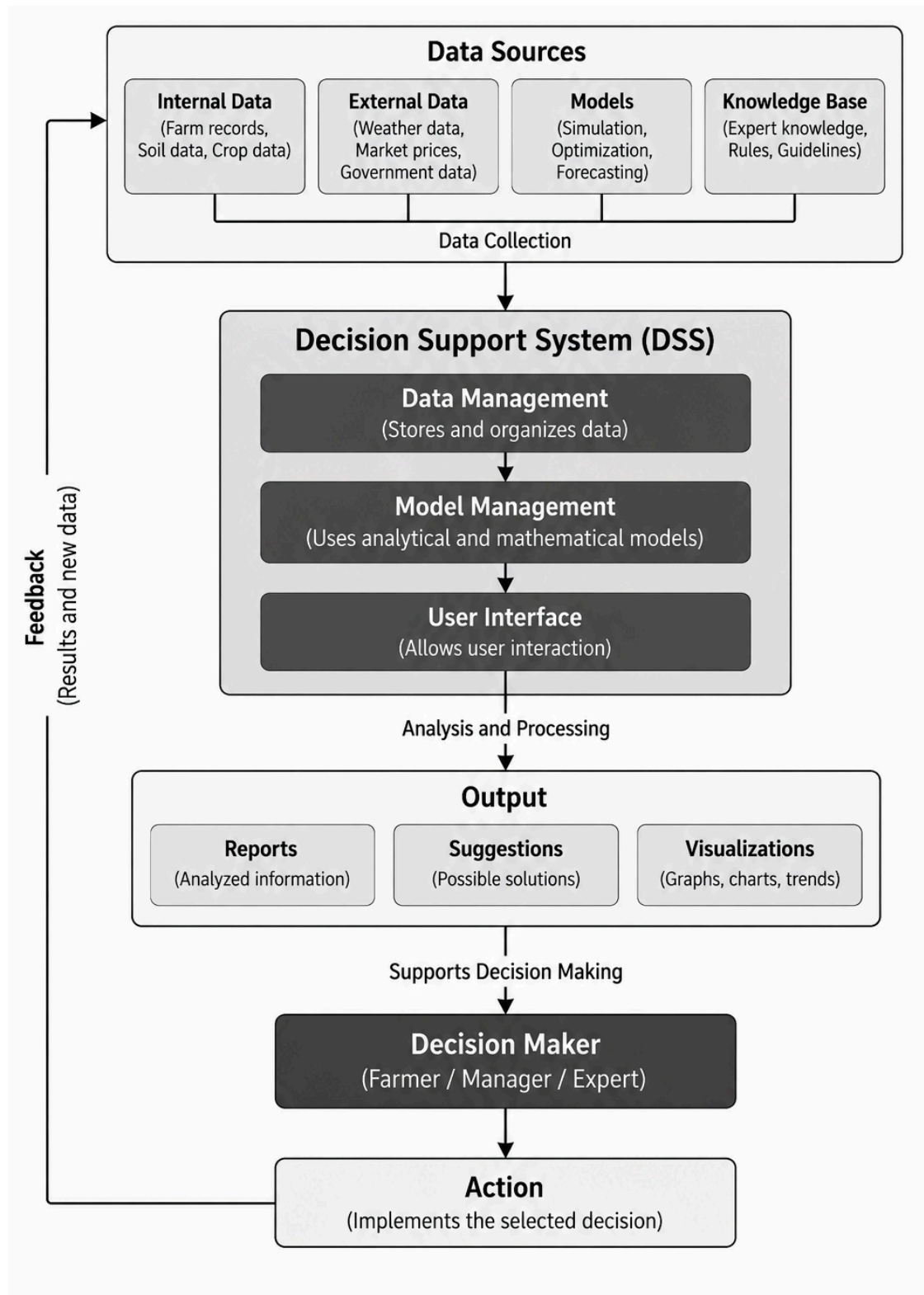
## **Examples of Agri E-Commerce Models**

| Model                                                                               |                     | Main Purpose                          |
|-------------------------------------------------------------------------------------|---------------------|---------------------------------------|
|    | Online Marketplace  | Farmers sell products online          |
|    | Input E-Commerce    | Buy seeds, fertilizers and pesticides |
|  | Farm-to-Table       | Direct delivery of farm products      |
|  | Digital Cooperative | Farmers sell together online          |
|  | B2B Marketplace     | Farmers sell in bulk to businesses    |
|  | Agri-Fintech        | Digital loans, insurance and payments |

# Expert and Decision Support System (DSS)

A Decision Support System (DSS) is a computer-based system that helps people analyze data and make informed decisions.

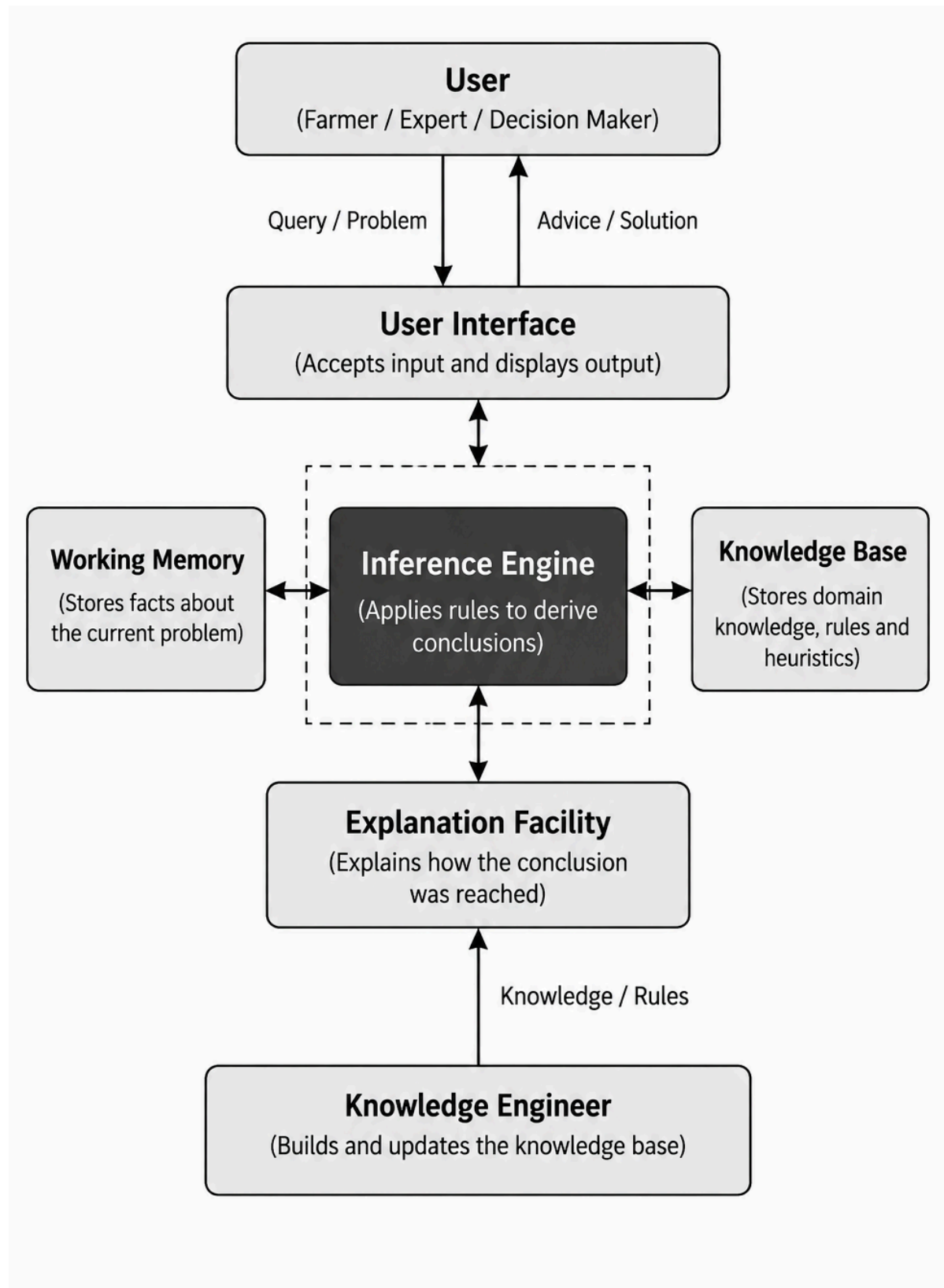
## DSS Workflow



## Expert Systems

An **expert system** is an AI system that works like a **human expert in a specific area**.

### **Example: Crop disease expert system**



## **Applications of Decision Support Systems**

The study material lists the following major applications:

1. **Diagnostics** – finding crop problems.
2. **Marketing** – helping with agricultural marketing decisions.
3. **Systems operations management** – managing agricultural operations.
4. **Automatic control** – controlling systems automatically.
5. **Strategic planning** – supporting long-term planning.